GENERAL EMBRYOLOGY.

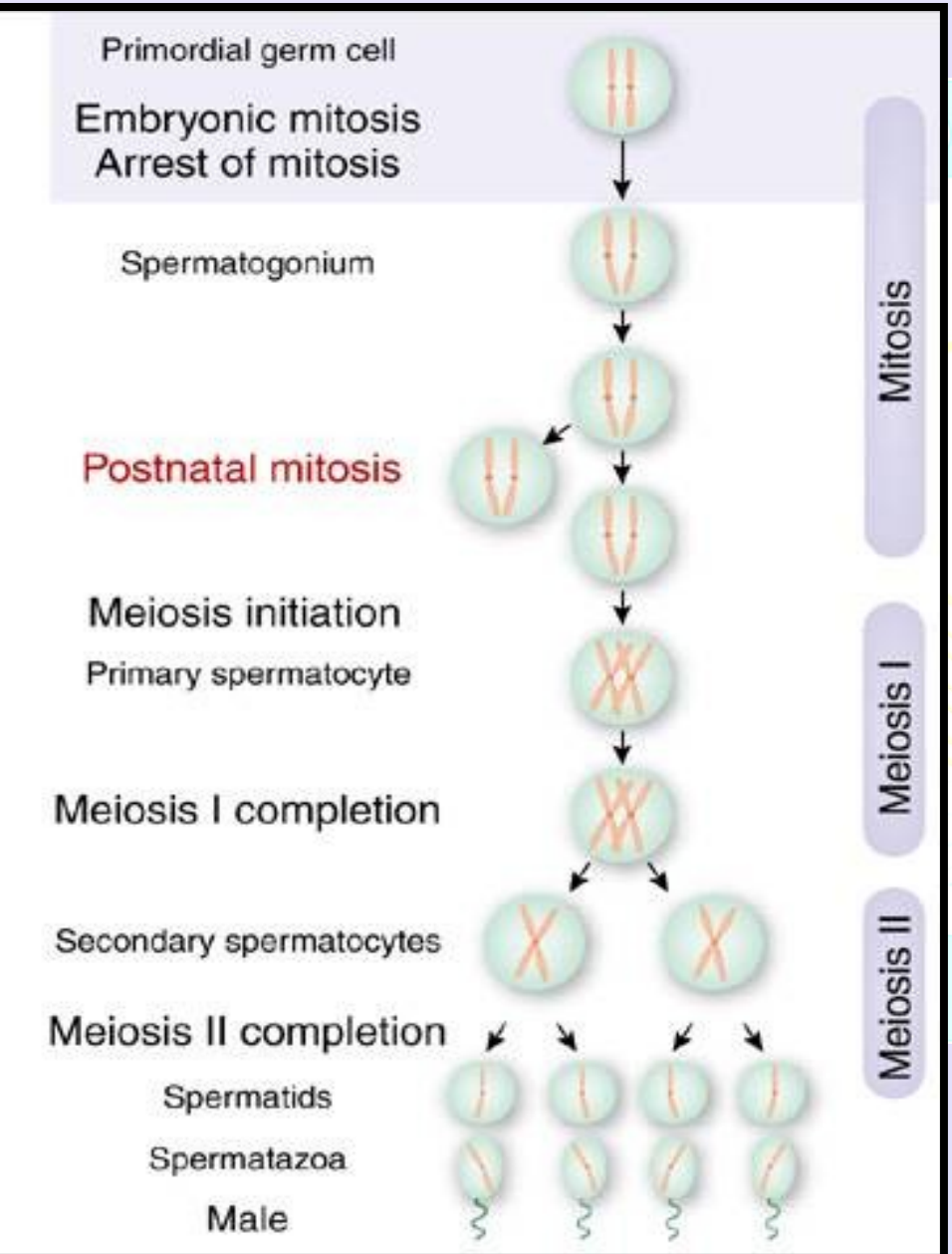
GAMETOGENESIS SPERMATOGENESIS AND SPERMIOGENESIS.

LEARNING OBJECTIVES

By the end of the lecture, the student will be able to :

- **Define gametogenesis.**
- **Describe the sequence events of spermatogenesis in the male.**
- **Identify the importance of mitosis & meiosis in spermatogenesis.**
- **List the steps in spermiogenesis.**
- **Differentiate between spermatogenesis & spermiogenesis.**

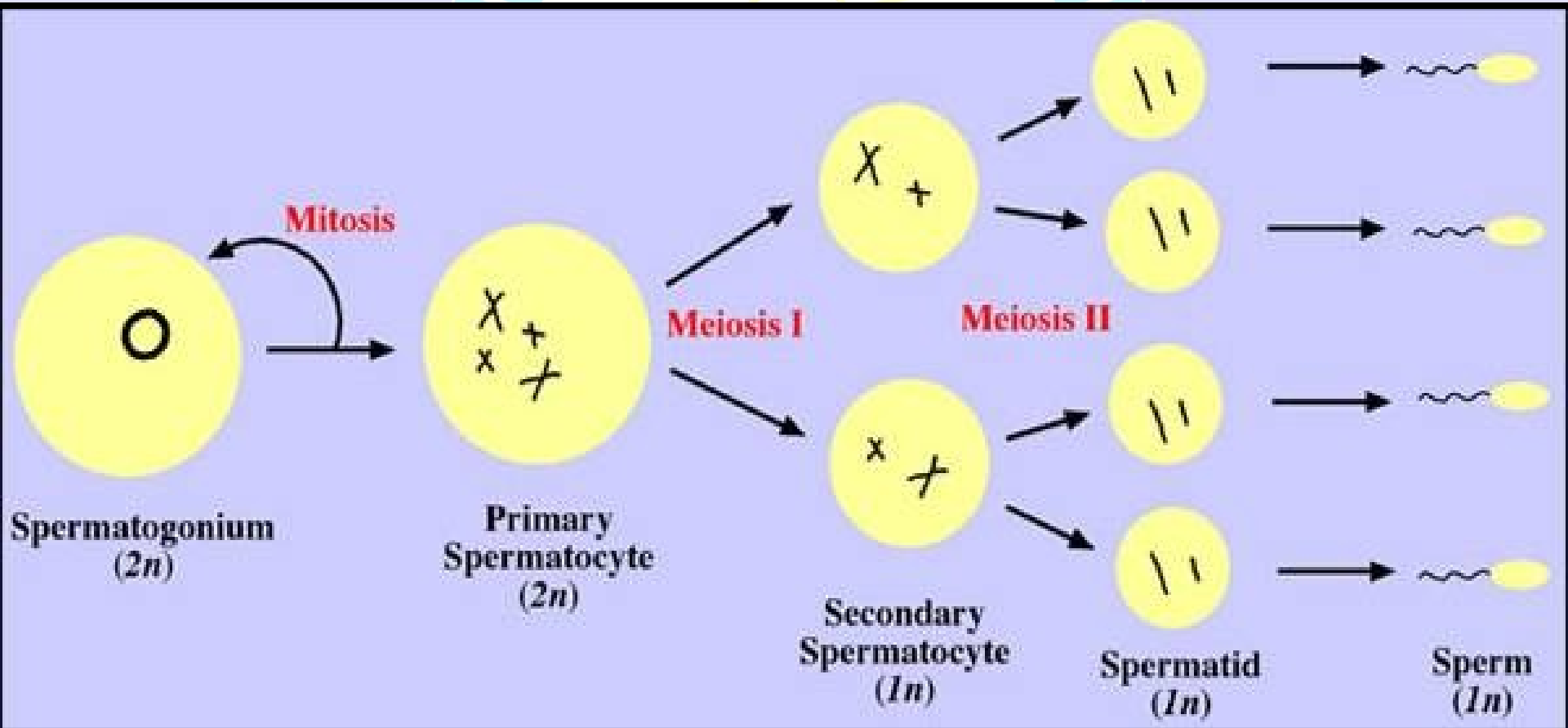
GAMETOGENESIS



- **This is the process of formation and development of specialized germinative cells, the gametes.**

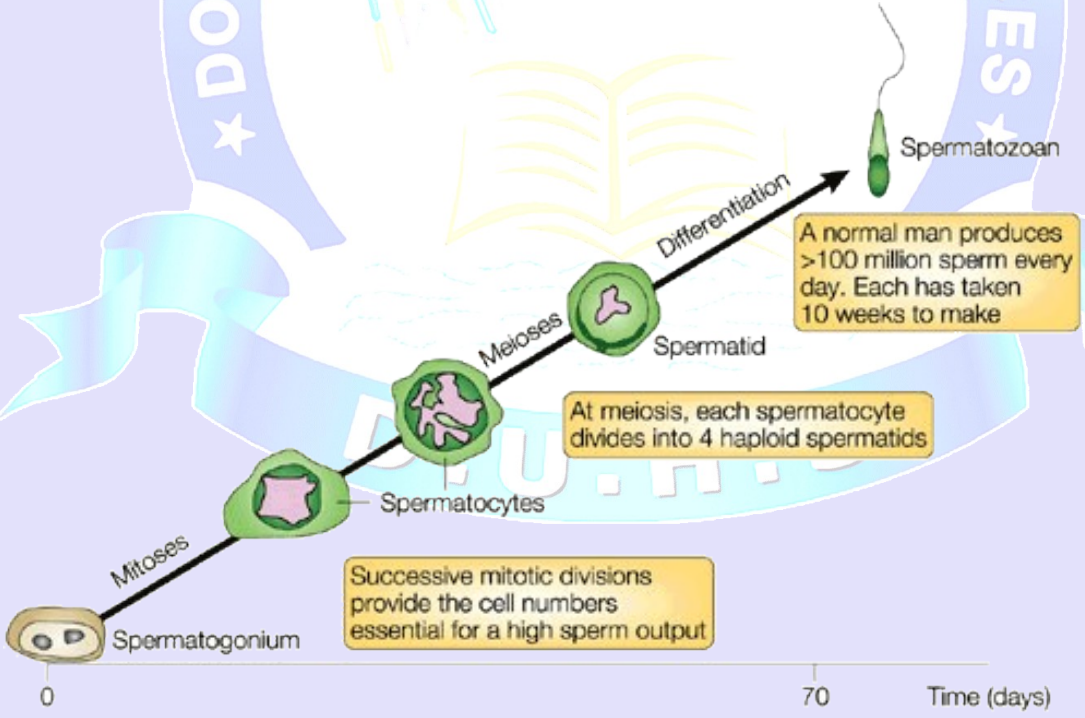
SPERMATOGENESIS

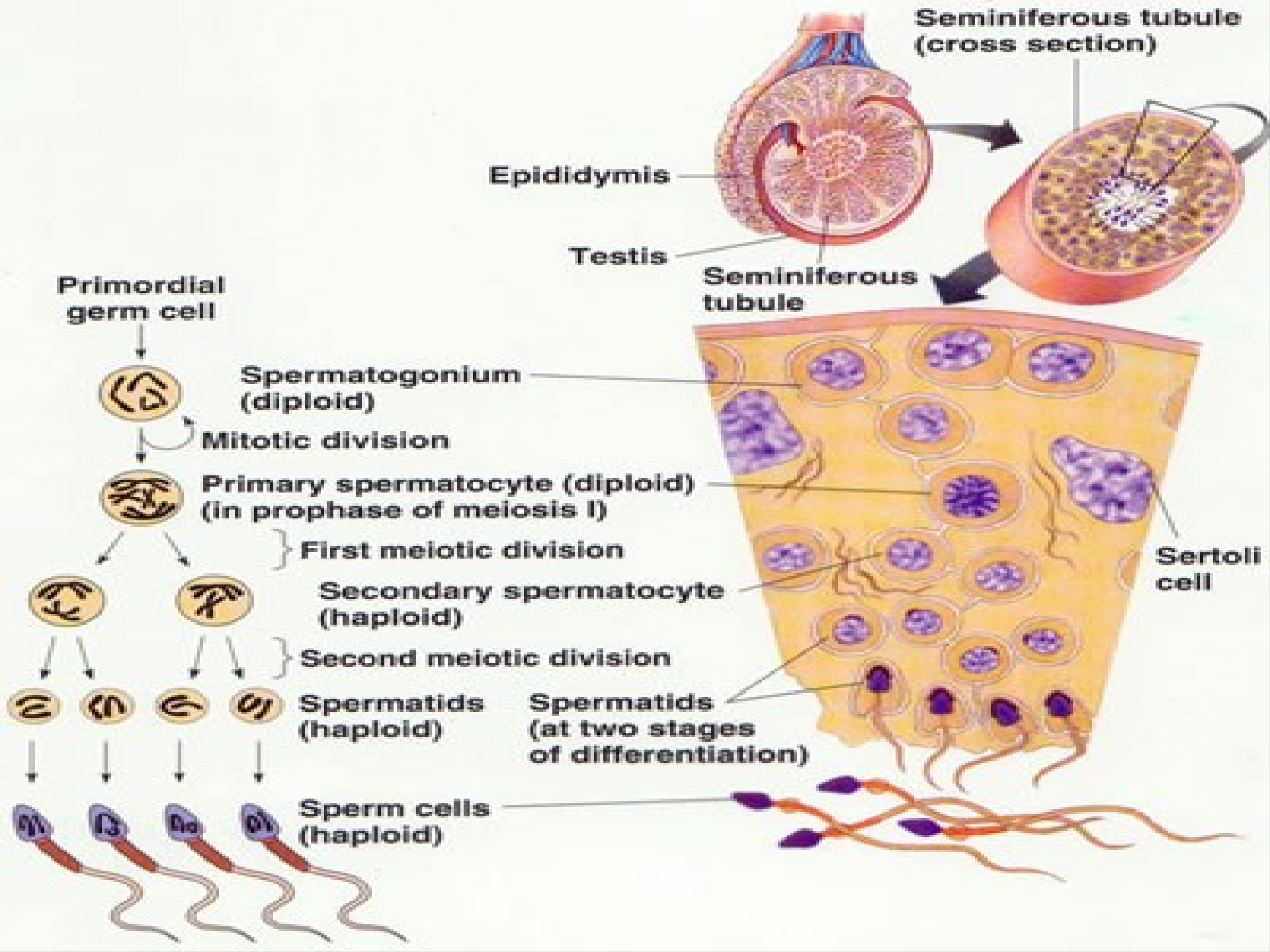
- Sermatogenesis is the sequence of events by which spermatogonia are transformed into mature sperms



Spermatogenesis

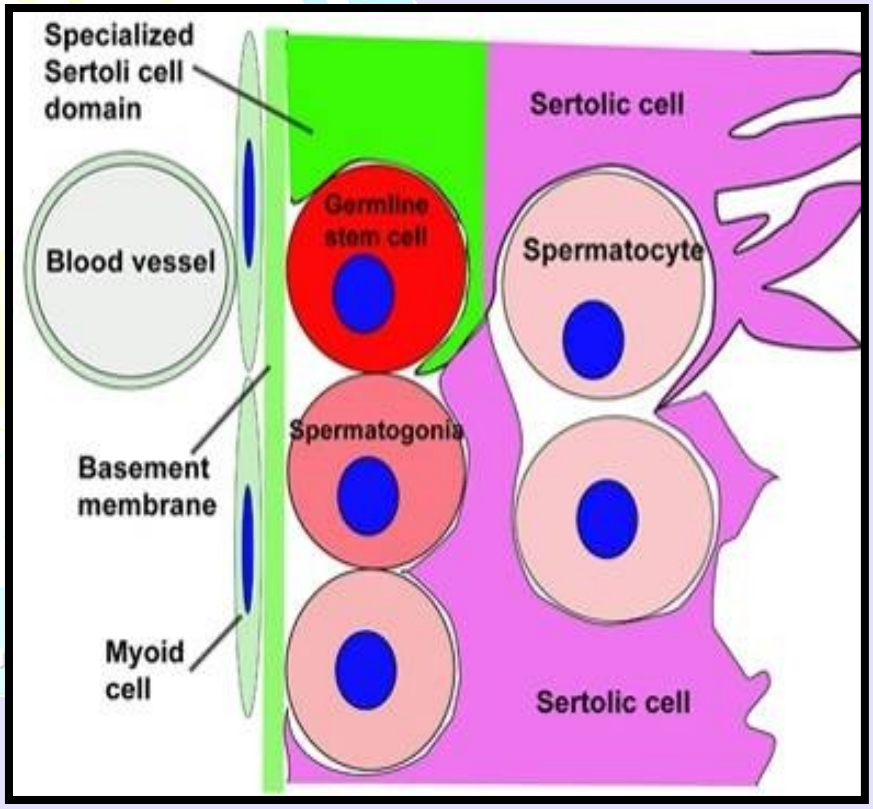
- Differentiation of **primordial germ cells** in the male begins at puberty.
- In the female, it begins in utero (up to Primary oocyte) during the 3rd month of development.

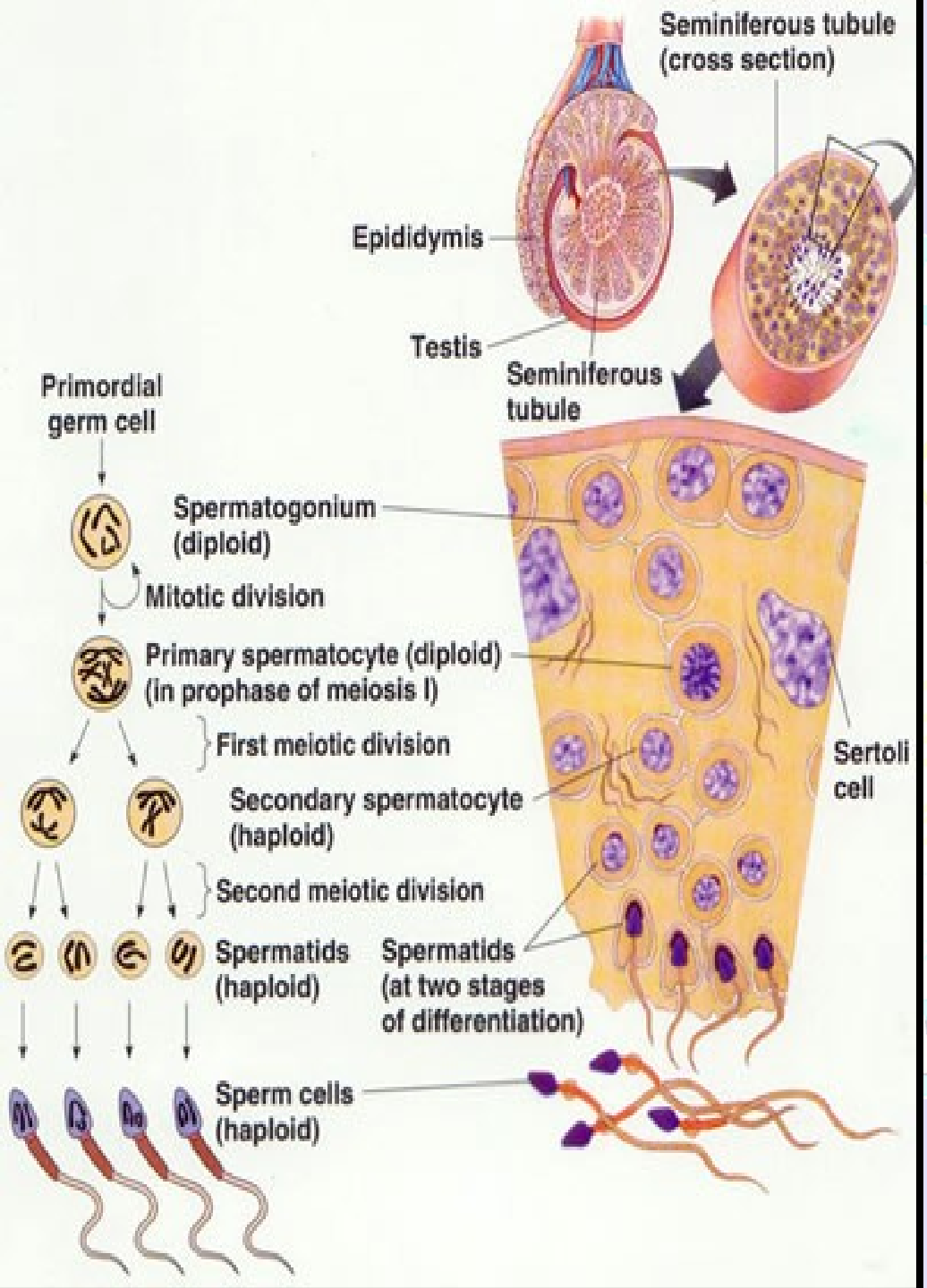



Seminiferous tubule
(cross section)
Epididymis
Testis
Seminiferous
tubule
Primordial
germ cell
Spermatogonium
(diploid)
Mitotic division
Primary spermatocyte (diploid)
(in prophase of meiosis I)
First meiotic division
Secondary spermatocyte
(haploid)
Second meiotic division
Spermatids
(haploid)
Spermatids
(at two stages
of differentiation)
Sertoli
cell
Sperm cells
(haploid)

SPERMATOGENESIS

- **Germ cells** in the male at birth in the sex cords of the testis are large, pale cells surrounded by supporting cells .
- Supporting cells become **sustentacular cells** or **sertoli cells**.

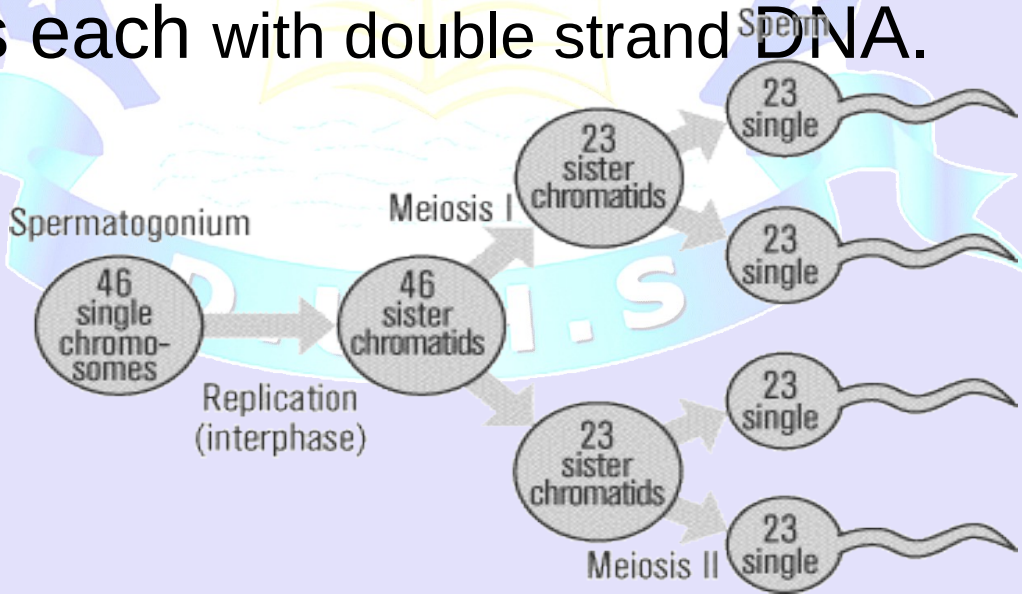




At puberty the sex chords acquired the lumen and differentiate into the **seminiferous tubules.** And primordial germ cells divide several times by Mitosis differentiating into **spermatogonia**.

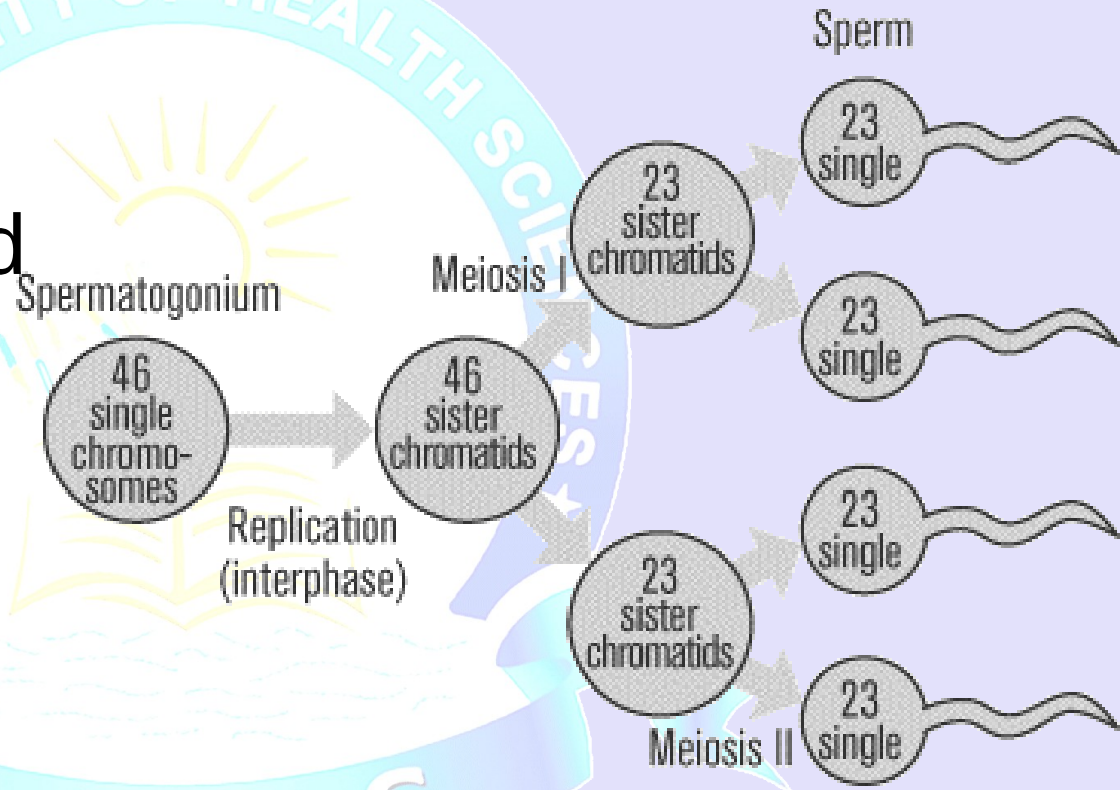
SPERMATOGENESIS

Spermatogonia divide repeatedly, via mitotic division to produce further spermatgonia of type **A** and type **B**. **Type A** spermatogonia are the reserve cells while **Type B** further differentiated either Type **B** Spermatogonia or ***Primary spermatocytes,*** still with 46 chromosomes. These divided by meiosis I to produce **secondary spermatocytes,** with 23 chromosomes each with double strand DNA.



SPERMATOGENESIS

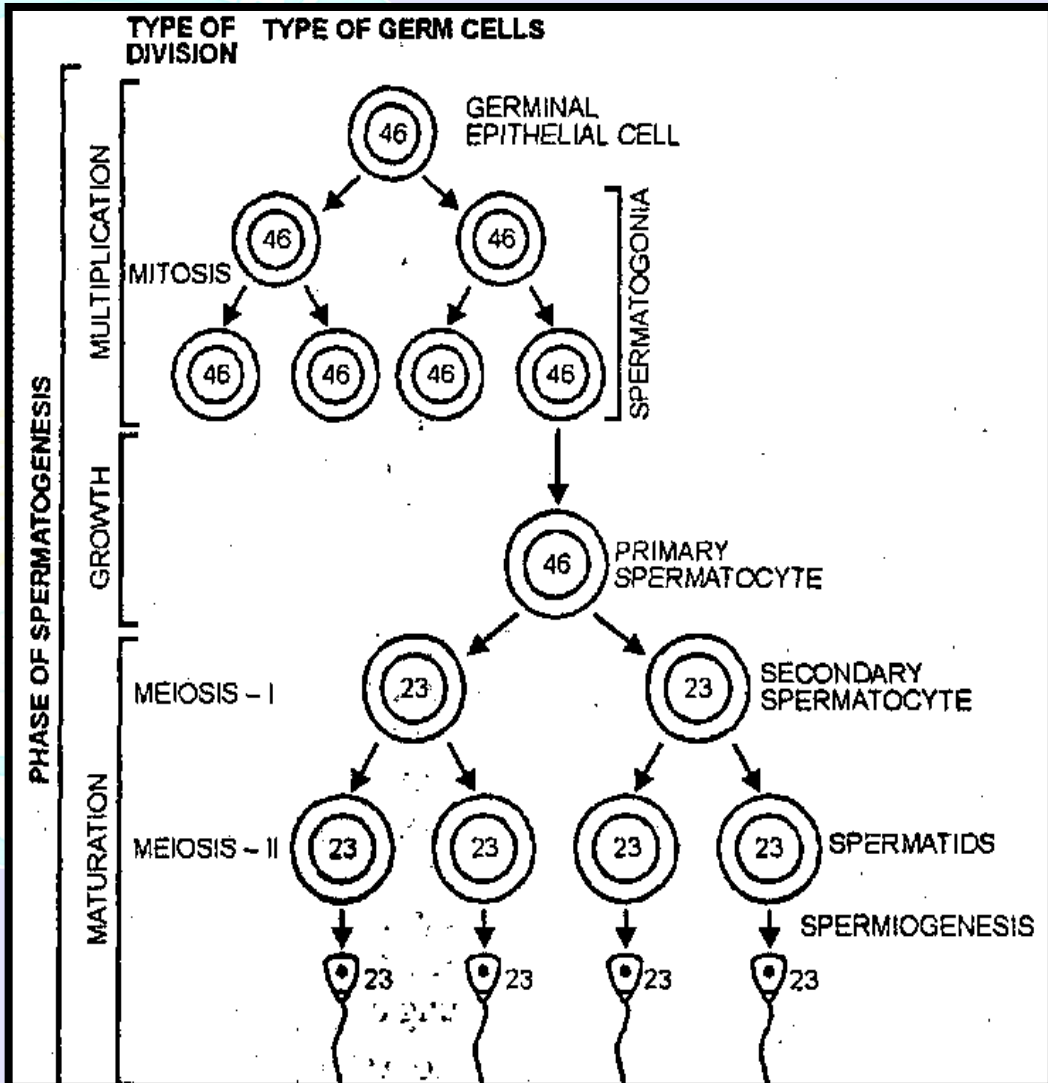
- The 2nd meiotic division produces **spermatids,** with **23n** & single strand **DNA** which then develop into **spermatozoa, or sperm cells.** These sperm cells have 23 chromosomes each, half the number needed to initiate human



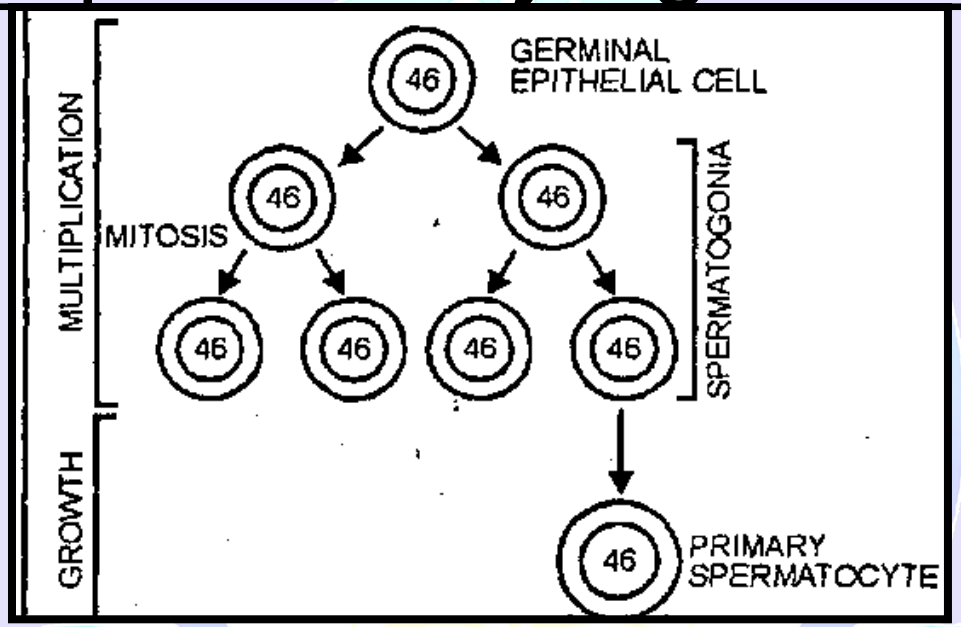
PHASES OF SPERMATOGENESIS

Process of spermatogenesis is divided into following phases:

- **M**ultiplication phase
- **G**rowth phase
- **M**aturation phase
- **S**permiogenesis



Multiplication Phase (Spermatocytogenesis).



- Sperm mother cells present in germinal epithelium of seminiferous tubules divide repeatedly by mitosis to form large number of diploid rounded sperm mother cells which are called as **spermatogonia.**
- Some of these sex cells move towards the lumen of seminiferous tubules and enter the growth phase. These cells are called **primary spermatocytes.**

MULTIPLICATION PHASE (SPERMATOCYTOGENESIS).

- The **primary spermatocytes** are diploid and contain (44 + XY) chromosomes.
- Some of the sex cells produced by the division of spermatogonia remain in their original condition (Spermatogonium A) and continue to divide giving rise to primary spermatocytes.
- Such cells are known as **stem cells (Spermatogonium A) .**

GROWTH PHASE

During this phase the spermatocyte as well as its nucleus enlarges in size. It gets ready to undergo maturation division.

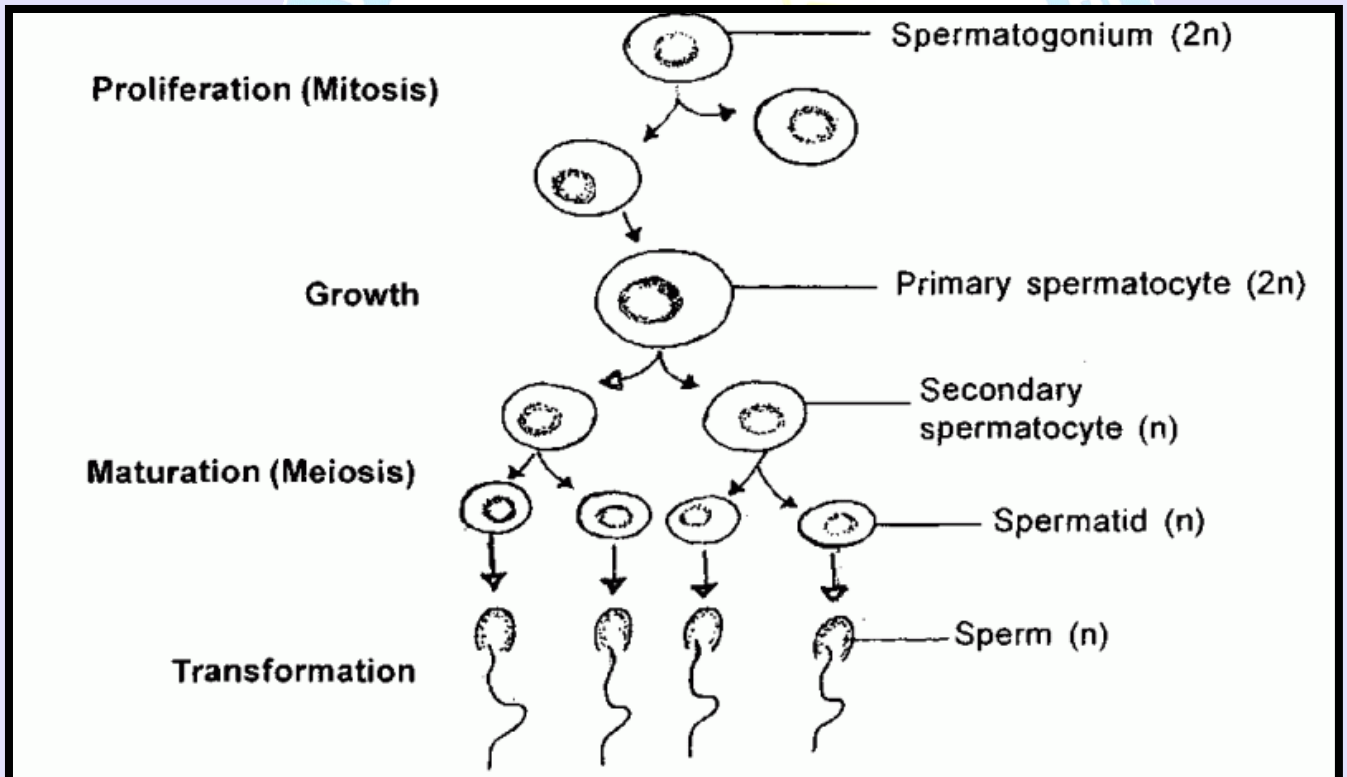


Fig 3.7 Stages in spermatogenesis

Maturation Phase

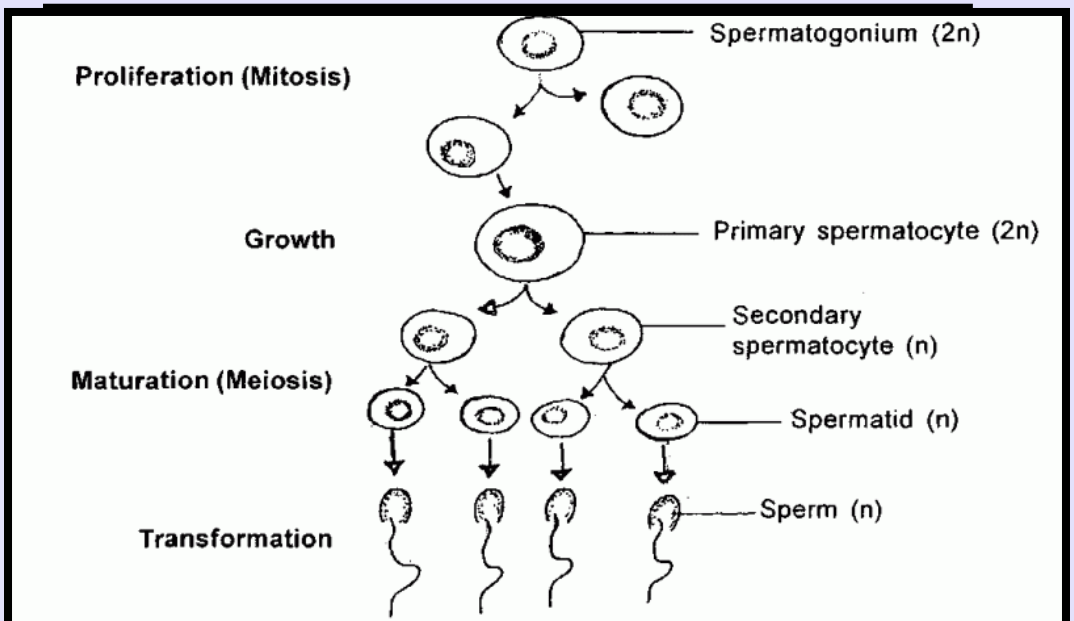


Fig 3.7 Stages in spermatogenesis

- Each diploid primary spermatocyte undergoes **meiosis I,** which is a reduction division.
- Two daughter cells are formed each with **'n' number of chromosomes & Double chain DNA.**
- The daughter cells are called **secondary spermaotcytes.**

MATURATION PHASE

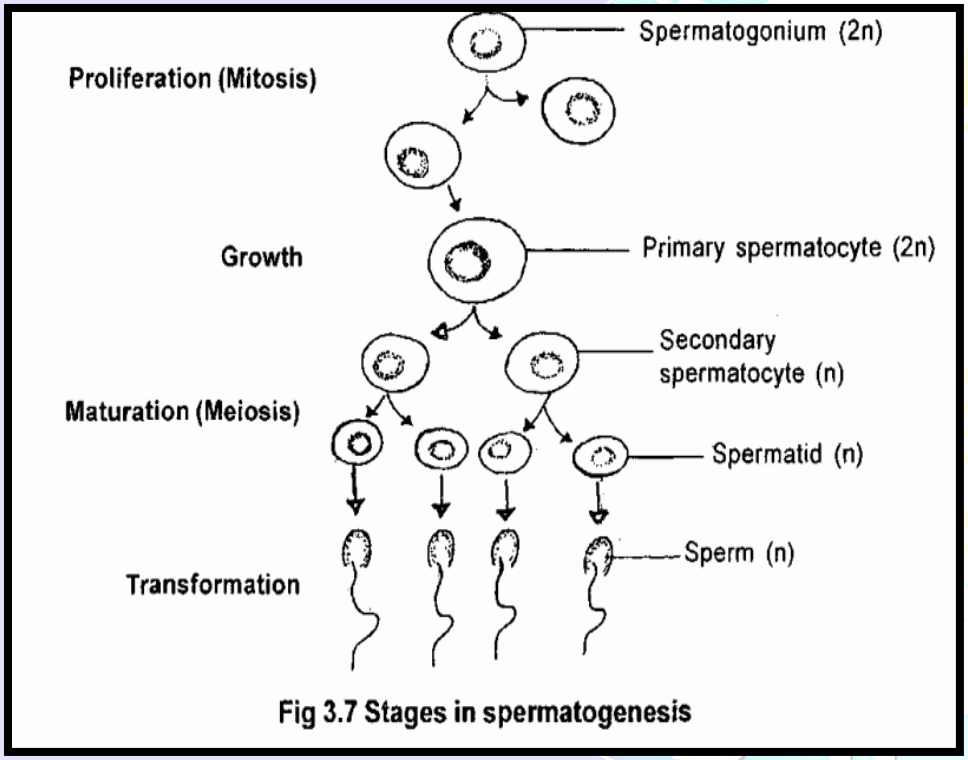


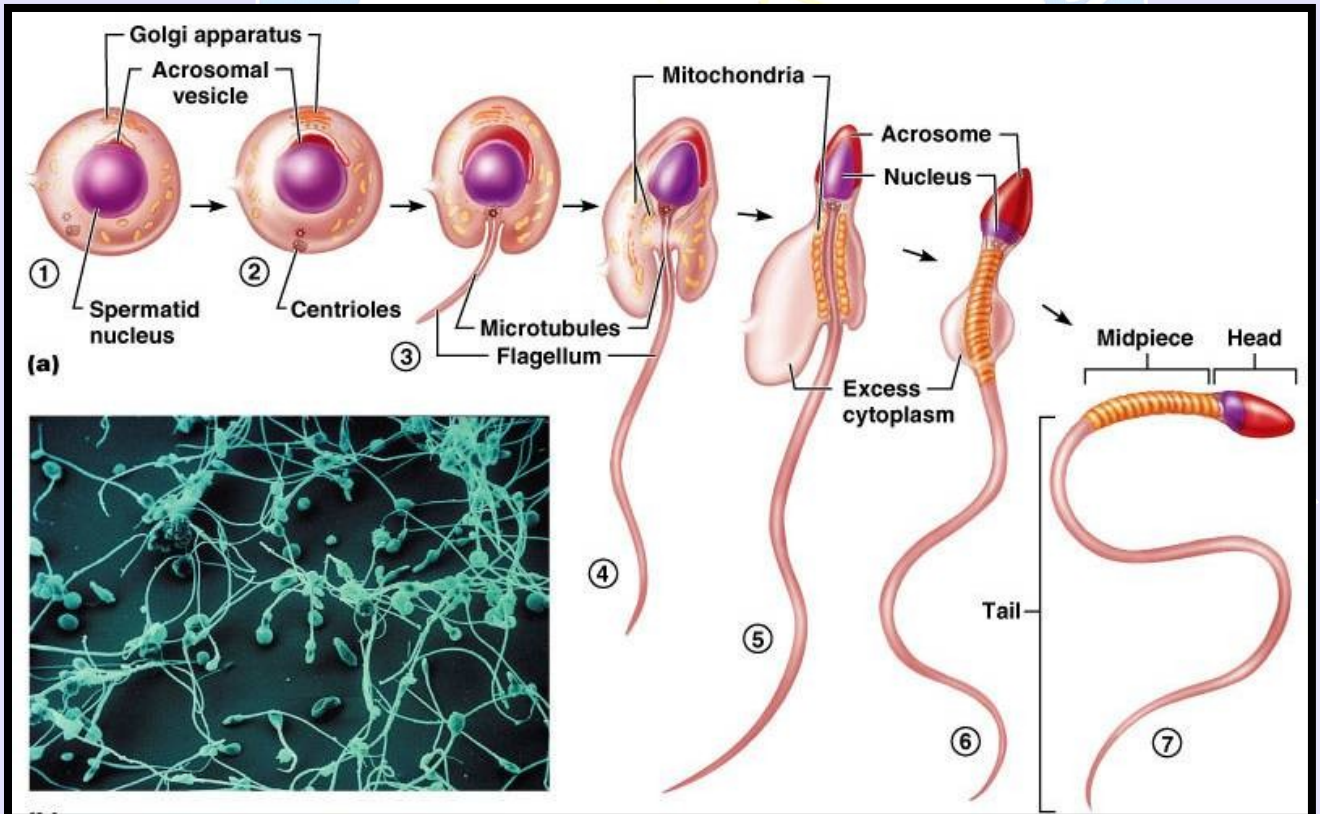
Fig 3.7 Stages in spermatogenesis

- The secondary spermatocytes are haploid and much smaller comparatively, containing (22+X) or (22+Y) chromosomes.

The secondary spermaotcytes undergo the second meiotic division which is similar like mitosis in which there is breakage of Double Chain of DNA resulted in 23n Single chain chromosome. This results in the formation of four daughter cells known as spermatids.

SPERMIOGENESIS

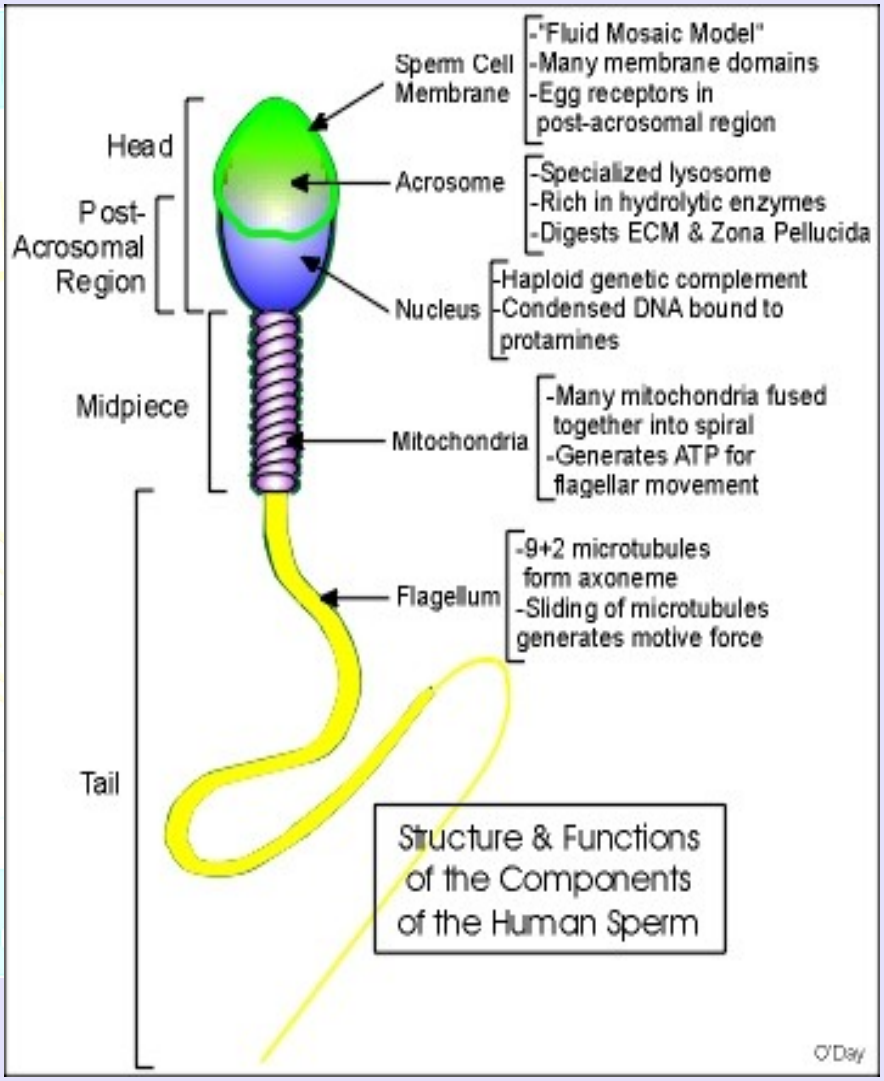
- **Spermiogenesis** is the final stage of spermatogenesis, which results in the maturation of spermatids into mature, spermatozoa.



Steps in SPERMIOGENESIS

a) Formation of an acrosome:

- **Acrosome** covers anterior 2/3 of nucleus and contains enzymes to assist in penetration of the egg and its surrounding layers during penetration.

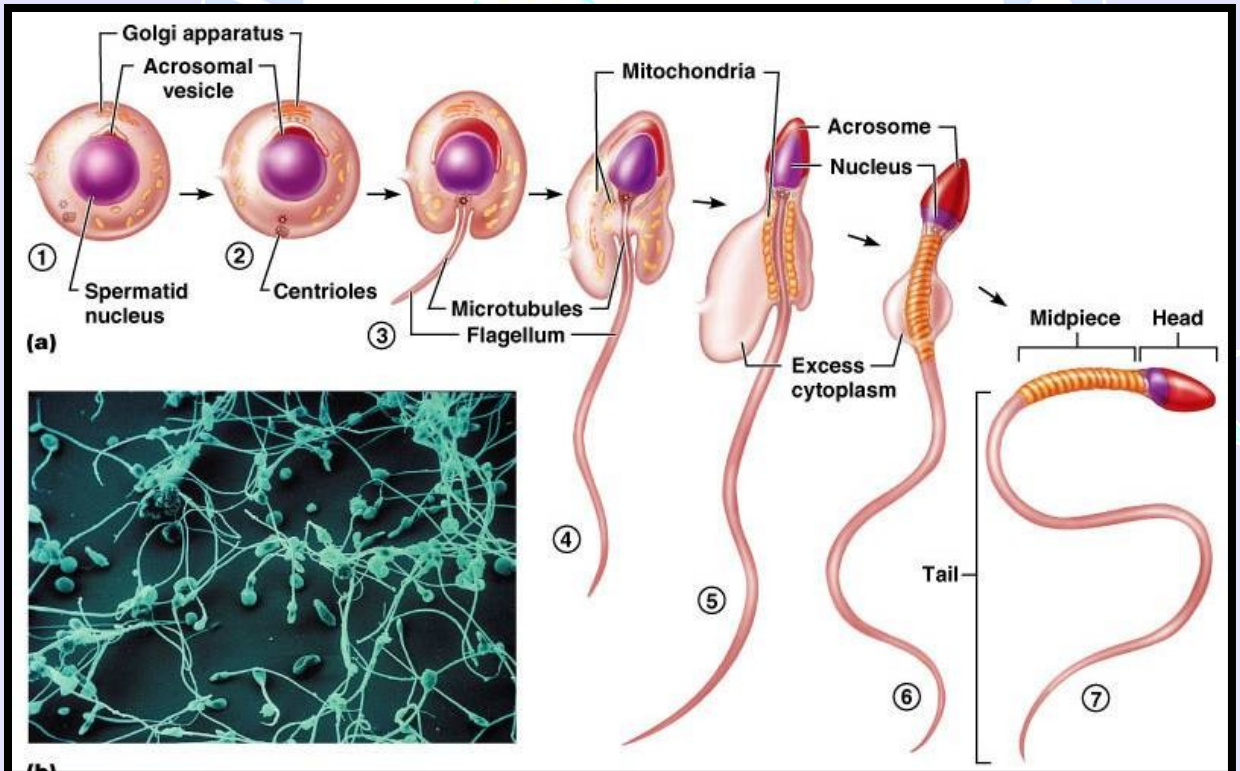


SPERMIOGENESIS

b) **C**ondensation of the **n**ucleus.

c) **F**ormation of the neck, middle piece and tail.

d) **S**hedding of most of the cytoplasm.

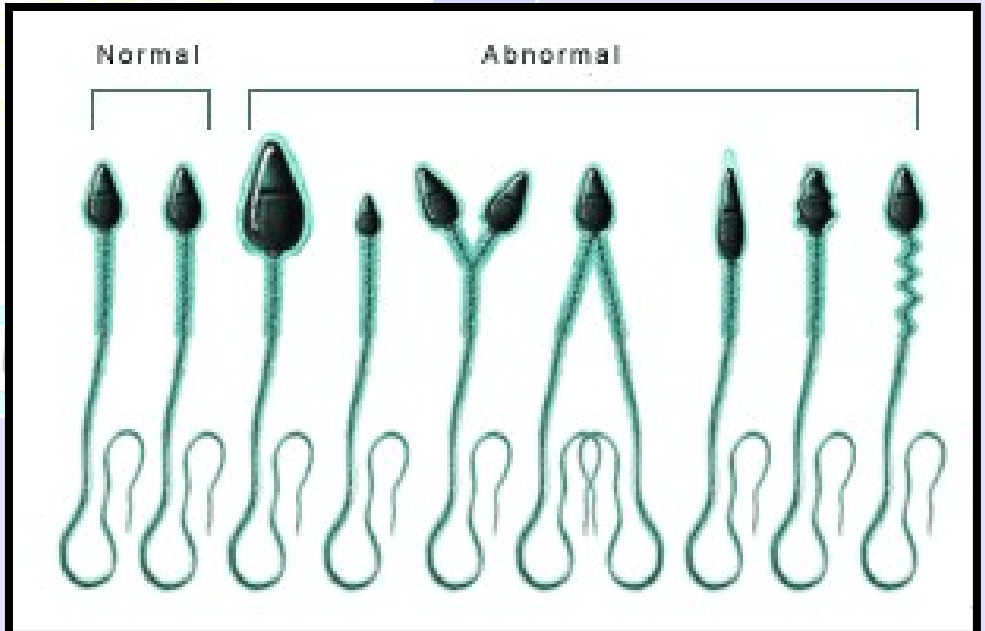


SPERMIOGENESIS

- Time required for a spermatogonium to develop into a mature spermatozoon is approximately 74 days.
- Fully developed spermatozoa enter the lumen of seminiferous tubules.
- Spermatozoa are pushed towards the epididymis by contractile elements in the wall of the seminiferous tubules.
- Initially only slightly motile, spermatozoa obtain full motility in the Cauda epididymis.

CLINICAL CORRELATES

- Abnormal spermatozoa are seen frequently.
- Upto 10% of all spermatozoa have observable defects.
  - **Head and tail may be abnormal.**
  - **may be giants or dwarfs.**
  - **May be joined.**



REFERENCES

- **Keith L. Moore Developing Human**

  8th Edition

  Chapter 2

  Pages 17- 19



THANK YOU